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Paper Code : 21314

F-414

B.C.A. (Third Semester)

Examination, 2023-24

(New Course)

Paper - BCA-304-N

OBJECT ORIENTED DESIGN AND PROGRAMMING IN C++

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) What is an operator? How many types of operators are used in C++. 7

   (b) Write a program in C++ to find the greatest number among three number. 7

2. (a) What is the difference between procedure oriented and object oriented programming. 7

   (b) Explain data abstraction, data encapsulation and data hiding. 7

3. (a) Explain switch case statement with the help of program. 7

   (b) What is the difference between structure and class explain it. 7

4. (a) Explain function overloading in detail. 7

   (b) What is the need of friend function? Explain with the help of program. 7

5. (a) Write a program of constructor over-loading. 7

   (b) Write a program of class which contains member function and define inside and outside definition. 7

6. (a) What is the difference between call by value and call by reference with the help of program. 7

(b) What is scope resolution operators reference variable? Also write their uses and their syntax. 7

7. (a) What is inheritance? How many types of inheritance are there? Also explain different type of derivation. 7

(b) Write a program of multiple inheritance. 7

8. (a) What is polymorphism? How many types of polymorphism exist? Also write their differences. 7

(b) What is virtual function? Write their characteristics and syntax. 7

Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21314

F-414

B.C.A. (Third Semester) Examination, 2021-22

(New Course)

Paper No. BCA-304-N

OBJECT ORIENTED PROGRAMMING USING C++

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) What are the elements of Object Oriented Programming? 7

   (b) What are the various loop control statements in C++? Explain briefly. 7

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2. (a) What are objects? Describe the syntax of defining objects with example. 7

   (b) List the various restrictions with operator overloading. Write a C++ program to illustrate the overloading of increment operator. 7

3. (a) Define a friend function. Explain the relation of friend function with respect to public, private and protected data members of the class. 7

   (b) Differentiate between private and protected class. Elaborate with suitable example. 7

4. (a) Write a C++ program to calculate the length of entered string. 7

   (b) Define a class template. What are its advantages? 7

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5. (a) Explain 'dynamic binding' with the help of an example. Give its advantages and disadvantages. 7

(b) Justify the need of virtual function in C++. 7

6. (a) What are the base and derived classes? Give a suitable example of inheritance. 7

(b) What are constructors and destructors? Explain how they differ from normal functions. https://www.mjpruonline.com

7. (a) Define pointers. How the arguments are passed in a function using pointers in C++. 7

(b) Write a C++ program to find the factorial of entered number using recursion. 7

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8. Write short notes on any **four** of the following: 3.5×4=14

(i) Inline Functions

(ii) Variables and Constants

(iii) Scope Resolution Operator

(iv) Library Functions

(v) Data Abstraction and Encapsulation

(vi) Break and Continue Statement.

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Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21314

F-414

B.C.A. (Third Semester) Examination, 2021

(New Course)

Paper - BCA-304-N

OBJECT ORIENTED DESIGN AND PROGRAMMING IN C++

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any five questions. All questions carry equal marks.

1. (a) How object oriented programming languages is different from procedural oriented languages?

   (b) What do you mean by "Class" and "Object"? Describe using a small code.

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2. (a) How to achiene abstraction?

   (b) Describe OOD-Methodology. What is SA/SD?

3. (a) Write a program using "class" and "object" to perform all basic arithmetic operations.

   (b) Explain various operators in C++.

4. (a) Differentiate "continue" and "break" statements with the help of a code.

   (b) How "Call-by-value" is different from "Call-by-Reference?

5. (a) How "Scope resolution operator" is useful in C++ programming? Justify using a code.

   (b) What is "Friend-Function"?

6. (a) Elaborate the advantages of "Inheritence" in any programming. What are its different types? Explain using their structures.

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(b) Write differences between compile time polymorphism and run-time polymorphism clearly.

7. (a) Write a program to show the concept of operator overloading.

(b) What are "virtual functions"?

8. Write short notes on the following :

(a) Pointers in C++

(b) JSD

(c) Metadata

(d) Static Binding versus Dynamic Binding

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Paper Code : 21314

F–414

B.C.A. (Third Semester) Examination, 2018

(New Course)

Paper-BCA-304-N

OBJECT ORIENTED DESIGN AND PROGRAMMING IN C++

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks.

1. (a) Define the characteristics of OOP's.

   (b) Define the various visibility modes used in C++ and their purpose also.

2. (a) Explain Storage classes.

   (b) What is the difference between Procedure Oriented Programming and OOP's ?

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3. (a) What do you mean by link and association ?

   (b) What is friend function ? Write its characteristics. Explain with the help of program.

4. (a) What is the difference between break and continue ? Explain each with program.

   (b) What is the recursive function ? Explain with the help of program.

5. (a) What is constructor and destructor ? Write its characteristics.

   (b) Write a program enter 10 number in array and print the number of array in reverse order.

6. (a) Explain the term runtime polymorphism and compile time polymorphism.

   (b) What is the difference between classes and structure ?

7. (a) What is operator overloading ? Explain in detail.

   (b) Explain the single inheritance with proper program.

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8. Write short notes on the following terms :

   Inline function, Switch statement, Metadata, Application of OOP's.

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